

MT 511: Sensor Materials and Technologies

Course Code	MT 511
Course Name	Sensor Materials and Technologies
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech./M.Tech./M.Sc./Ph.D.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction and Fundamental Concepts:** Overview of sensors, materials used for sensors, multidisciplinary aspects of sensor technology, key sensor parameters such as sensitivity, range, accuracy, and resolution. (5 hours)
- **Sensor Materials:** Examination of sensor materials including silicon, plastics, metals, ceramics, structural glasses, optical glasses, and nanomaterials. (4 hours)
- **Physical Principles of Sensing:** Study of sensing principles for various physical phenomena, including force, temperature, vibration, pressure, flow, and optical properties. This section also explores a range of sensors such as piezoelectric, electrical, acoustic, pneumatic, magnetic, optical, and pH sensors. (7 hours)
- **Sensor Fabrication and Characterization:** Overview of fabrication processes such as surface processing (spin casting, vacuum deposition, sputtering, chemical vapor deposition, electroplating). Introduction to Microelectromechanical Systems (MEMS) technologies, including lithography, silicon micromachining, micromachining of bridges and cantilevers, lift-off, wafer bonding, and LIGA technologies. Fundamentals of etching techniques (both wet and dry), and microscopy techniques (optical and electron microscopy, among others). (12 hours)
- **Signal Conditioning, Processing, and Display Systems:** Covers signal conditioning methods like deflection bridges, amplifiers, A.C. carrier systems, current transmitters, oscillators, and resonators. Signal processing techniques such as analog-to-digital conversion, computer and microcontroller systems, and related software. Discusses data display methods, including pointer-scale indicators, LED displays, cathode ray tubes (CRT), liquid crystal displays (LCDs), and electroluminescence (EL) displays. (12 hours)
- **Sensor Applications in Various Fields:** Exploration of sensor applications, covering occupancy and motion detectors, position and displacement sensors, velocity, and acceleration sensors, as well as force, strain, tactile, pressure, and temperature sensors. Specific applications include sensors utilized in CNC machine tools (linear and angular position, velocity sensors) and acoustic emission, accompanied by an introduction to concepts in pattern recognition. (4 hours)

Laboratory Modules

1. NA

Text Books

1. John P. Bentley, **Principles of Measurement Systems**, 4th Edition, Pearson Education, 2005.
2. S.M. Sze, **Semiconductor sensors**, 3rd Edition, John Wiley & Sons Inc., 2006.
3. J. Fraden, **Handbook of Modern Sensors: Physical, Designs, and Applications**, AIP Press, Springer.
4. Neil W. Ashcroft and N. David Mermin, **Solid State Physics**, Saunders College Publishing, 1976.

References

1. NA